Substitution method

1 Solve the following simultaneous equations:

a $y = 6x + 56$
$y = 2x + 24$

b $y = -5x - 22$
$y = 7x + 38$

c $y = 5x + 37$
$5y = 35x + 265$

d $y = -4x - 27$
$2y = 10x + 54$

e $y = 3x + 14$
$x + y = 2$

f $y = -3x + 22$
$x + y = 10$

g $y = -6x + 59$
$-2x + y = -13$

h $y = 2x + 13$
$x - y = -11$

i $y = -3x + 9$
$6x - y = 0$

j $y = 4x - 28$
$3x + 5y = 44$

k $y = -3x - 23$
$x + 4y = -37$

l $y = 2x - 15$
$3x - 4y = 20$

m $5x - 2y = -10$
$y = -5x + 3$

n $2x + 3y = 26$
$8x - 5y = 36$

2 Consider the given system of equations:

$$3x - 7y = 4$$
$$-12x + 28y = -16$$

a Rearrange $3x - 7y = 4$ to find x in terms of y.

b Substitute your expression for x into $-12x + 28y = -16$ and solve for y.

c State whether the system of equations is inconsistent, dependent or independent.

3 Consider the given system of equations:

$$5x + 3y = -64$$
$$7x + 4y = -88$$

a Rearrange $5x + 3y = -64$ to make y the subject in terms of x.

b Solve for x.

c Hence, solve for y.

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4 Solve the following systems of equations:

a $\begin{aligned} 2x+3y&=26 \\ 8x-5y&=36 \end{aligned}$

b $\begin{aligned} 2x+7y&=23 \\ 5x+6y&=0 \end{aligned}$

c $\begin{aligned} 5x+4y&=10 \\ 3x+7y&=-17 \end{aligned}$

d $\begin{aligned} 4x+3y&=17 \\ 9x-4y&=6 \end{aligned}$

e $\begin{aligned} -7f+2g&=-\frac{2}{5} \\ -21f+10g&=\frac{18}{5} \end{aligned}$

f $\begin{aligned} -7p-2q&=-\frac{99}{40} \\ -21p-10q&=-\frac{85}{8} \end{aligned}$

g $\begin{aligned} -4a+7b&=\frac{23}{2} \\ -20a+21b&=\frac{45}{2} \end{aligned}$

h $\begin{aligned} 4v-7t&=-\frac{143}{20} \\ 12v-35t&=-\frac{779}{20} \end{aligned}$

5 Describe how to check whether a given ordered pair is a solution of a system of equations.

6 Determine whether $(5,2)$ is a solution of the following system of equations:

$$\begin{aligned} x+y&=7 \\ x-y&=3 \end{aligned}$$

7 Determine whether $(4,17)$ is a solution of the following system of equations:

$$\begin{aligned} y&=6x-7 \\ 4x+3y&=67 \end{aligned}$$

8 Consider the following system of linear equations:

$$\begin{aligned} -6x-2y&=-28 \\ 2x+16y&=40 \\ 4x-2y&=12 \end{aligned}$$

a Find the values of x and y that satisfy the first two equations.

b Determine if this solution satisfies the third equation.

c Hence state whether the lines are concurrent.

Applications

9 The length of a rectangle measures 10 units more than the width, and the perimeter of the rectangle is 56 units. Let y be the width and x be the length of the rectangle.

a Consider that the length of the rectangle is 10 units more than the width to set up Equation 1.

b Consider that the perimeter of the rectangle is equal to 56 to set up Equation 2.

c Solve for y to find the width.

d Hence, solve for x to find the length.

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10 There are 22 members in a group and the men outnumber the women by 6. Let x and y be the number of women and men in the group respectively.

- **a** Consider that the men outnumber the women by 6 to set up Equation 1.
- **b** Consider that there are a total of 22 members in the group to set up Equation 2.
- **c** Solve for x to find the number of women in the group.
- **d** Hence, solve for y to find the number of men in the group.

11 A mother is currently 10 times older than her son. In two years time, she will be 7 times older than her son. Let x and y be the present ages of the son and mother respectively.

- **a** Set up two equations that describe the statements above.
- **b** Solve for x to find the son's current age.
- **c** Hence, solve for y to find the mother's current age.

12 A man is 5 times as old as his son. Four years ago the man was 9 times as old as his son. Let x and y be the ages of the man and his son respectively.

- **a** Set up two equations that describe the statements above.
- **b** Solve for y to find the age of the son.
- **c** Hence, solve for x to find the age of the man.

13 Valerie has $\$3000$ to invest, and wants to split it up between two accounts. Account A earns 6% annual interest and Account B earns 8% annual interest. Her target is to earn $\$204$ total interest from the two accounts in one year. Let x and y be the amounts, in dollars, that she invests in accounts A and B respectively.

- **a** Set up two equations that describe the statements above.
- **b** Solve for x and y.

14 Neil bought some fresh produce. He picked up 2 apples, and 3 kiwi fruit. The cost of Neil's shopping was $\$10.08$. Eileen went to the same shop and bought 5 apples and 7 kiwi fruit. The cost of Eileen's shopping was $\$24.08$. Let f and g be the price of apples and kiwi fruit respectively.

- **a** Set up two equations that describe the statements above.
- **b** Find the price of apples and kiwi fruit respectively.

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15 The number of new jobs created in Venhurst varies greatly each year. The number of jobs created in 2013 was $190\,000$ less than quadruple the number of jobs created in 2008. This is equivalent to an increase of $560\,000$ jobs created from 2008 to 2013. Let x be the number of jobs created in 2008 and let y be the number of jobs created in 2013.

a Set up two equations that describe the statements above.

b Find the amount of jobs created in 2008 and 2013.

16 A taxi driver charges $\$x$ for the initial pick up, and then $\$y$ per km travelled. A journey of 10km costs $\$105$, and a journey of 15km costs $\$155$.

a Set up two equations that describe the statements above.

b Find the amount of the fixed charge and the charge per km.

c How much does a person have to pay for traveling a distance of $25\ \text{km}$?

17 The coach of a tennis team buys 7 rackets and 6 balls for $\$3800$. Later, he buys 3 rackets and 5 balls for $\$1750$. Find the price of rackets and balls respectively.

18 Toby's piggy bank contains only $5c$ and $10c$ coins. If it contains 60 coins with a total value of $\$3.75$, find the number of each type of coin.

19 Michael sells hot dogs and soda. Each hotdog costs $\$1.50$ and each soda costs $\$0.50$. At the end of the day he made a total of $\$78.50$ for a total of 87 hot dogs and sodas combined. How many hotdogs and sodas were sold?

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